

MAT1001 — Midterm (Reference Solutions)

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Notes: This file contains reference solutions only (no question statements).**Question 1****(45 points)****Short answers (summary):**

- 1 (a)
- 2 (d)
- 3 vertical asymptote $x = 1$, oblique asymptote $y = x + 1$
- 4 True
- 5 True
- 6 $y = x - \pi - 9$
- 7 True
- 8 (d)
- 9 True
- 10 False
- 11 (d)
- 12 (b)
- 13 $\frac{1}{2}$
- 14 $\int_0^1 \frac{1}{1+x} dx$
- 15 $x^2 + \frac{2}{x} - \frac{\tan 2x}{2} + C$

Question 2**(24 points)****Final answer:** Method 1:

$$\begin{aligned}
\lim_{x \rightarrow 0} \frac{\sin(3x)}{\tan(2x)} &= \lim_{x \rightarrow 0} \frac{\sin 3x}{\sin 2x} \cdot \cos 2x = \lim_{x \rightarrow 0} \frac{\sin 3x}{\sin 2x} \cdot \lim_{x \rightarrow 0} \cos 2x \\
&= \lim_{x \rightarrow 0} \frac{\sin 3x/3x}{\sin 2x/2x} \cdot \frac{3x}{2x} \cdot \lim_{x \rightarrow 0} \cos 2x \\
&= \lim_{x \rightarrow 0} \frac{3x}{2x} \cdot 1 = \frac{3}{2}.
\end{aligned}$$

Method 2: By linearization,

$$(i) \quad \sin \theta = \theta + o(\theta), \quad \tan \theta = \theta + o(\theta), \quad \text{as } \theta \rightarrow 0.$$

Substituting $\theta = 3x$ and $\theta = 2x$ respectively, we have

$$\sin(3x) = 3x + o(x), \quad \tan(2x) = 2x + o(x), \quad \text{as } x \rightarrow 0.$$

Thus,

$$\lim_{x \rightarrow 0} \frac{\sin(3x)}{\tan(2x)} = \lim_{x \rightarrow 0} \frac{3x + o(x)}{2x + o(x)} = \lim_{x \rightarrow 0} \frac{3 + o(1)}{2 + o(1)} = \frac{3}{2}.$$

Method 3:

$$\begin{aligned}
\lim_{x \rightarrow 0} \frac{\sin(3x)}{\tan(2x)} &= \lim_{x \rightarrow 0} \frac{\frac{\sin 3x - \sin 3 \cdot 0}{x}}{\frac{\tan 2x - \tan 2 \cdot 0}{x}} = \frac{\lim_{x \rightarrow 0} \frac{\sin 3x - \sin 0}{x}}{\lim_{x \rightarrow 0} \frac{\tan 2x - \tan 0}{x}} \\
&= \frac{(\sin 3x)'|_{x=0}}{(\tan 2x)'|_{x=0}} = \frac{3 \cos 3 \cdot 0}{2 \sec^2 2 \cdot 0} = \frac{3}{2}.
\end{aligned}$$

Final answer:Method 1: Let $y = x - \pi$. Then $x = y + \pi$ and $y \rightarrow 0$ as $x \rightarrow \pi$. Thus,

$$\lim_{x \rightarrow \pi} \frac{\sin 3x}{\tan 2x} = \lim_{y \rightarrow 0} \frac{\sin 3(y + \pi)}{\tan 2(y + \pi)} = \lim_{y \rightarrow 0} \frac{\sin(3y + 3\pi)}{\tan(2y + 2\pi)} = \lim_{y \rightarrow 0} \frac{-\sin 3y}{\tan 2y} = -\frac{3}{2}.$$

Method 2: By linearization at $x = \pi$,

$$\begin{aligned}
(ii) \quad \sin 3x &= \sin 3\pi + 3 \cos 3\pi(x - \pi) + o(x - \pi) = -3(x - \pi) + o(x - \pi), \\
\tan 2x &= \tan 2\pi + 2 \sec^2 2\pi(x - \pi) + o(x - \pi) = 2(x - \pi) + o(x - \pi),
\end{aligned}$$

as $x \rightarrow \pi$. Thus,

$$\lim_{x \rightarrow \pi} \frac{\sin 3x}{\tan 2x} = \lim_{x \rightarrow \pi} \frac{-3(x - \pi) + o(x - \pi)}{2(x - \pi) + o(x - \pi)} = \lim_{x \rightarrow \pi} \frac{-3 + o(1)}{2 + o(1)} = -\frac{3}{2}.$$

Final answer:

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (x^2 - x - 6) = -4.$$

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} (x-2)^4 \sin \frac{1}{x-2}$$

Since $1 \leq \sin \frac{1}{x-2} \leq 1$, we have

(iii)

$$-(x-2)^4 \leq (x-2)^4 \sin \frac{1}{x-2} \leq (x-2)^4.$$

By Squeeze Theorem,

$$\lim_{x \rightarrow 2^-} (x-2)^4 \sin \frac{1}{x-2} = 0.$$

So $\lim_{x \rightarrow 2^-} f(x) = 0 \neq \lim_{x \rightarrow 2^+} f(x)$. Therefore, $\lim_{x \rightarrow 2} f(x)$ does not exist.

Final answer:

Method 1:

$$\lim_{x \rightarrow -\infty} (x + \sqrt{x^2 - x}) = \lim_{x \rightarrow -\infty} \frac{(x + \sqrt{x^2 - x})(x - \sqrt{x^2 - x})}{x - \sqrt{x^2 - x}}$$

$$= \lim_{x \rightarrow -\infty} \frac{x^2 - (x^2 - x)}{x - \sqrt{x^2 - x}} = \lim_{x \rightarrow -\infty} \frac{x}{x - \sqrt{x^2 - x}}$$

$$= \lim_{x \rightarrow -\infty} \frac{1}{1 + \sqrt{1 - \frac{1}{x}}} = \frac{1}{1+1} = \frac{1}{2}.$$

Method 2: $x + \sqrt{x^2 - x} = x - x\sqrt{1 - \frac{1}{x}}$ if $x < 0$. By linearization,

(iv)

$$(1 + \theta)^{\frac{1}{2}} = 1 + \frac{1}{2}\theta + o(\theta), \text{ as } \theta \rightarrow 0.$$

Substituting $\theta = -\frac{1}{x}$, we have

$$\sqrt{1 - \frac{1}{x}} = 1 - \frac{1}{2x} + o\left(\frac{1}{x}\right), \text{ as } x \rightarrow -\infty.$$

Thus,

$$x + \sqrt{x^2 - x} = x - x\left(1 - \frac{1}{2x} + o\left(\frac{1}{x}\right)\right) = \frac{1}{2} + o(1), \text{ as } x \rightarrow -\infty.$$

Therefore,

$$\lim_{x \rightarrow -\infty} (x + \sqrt{x^2 - x}) = \frac{1}{2}.$$

Question 3**(24 points)****Solution**

Final answer: Since $-1 \leq \sin \frac{1}{x-1} \leq 1$, we have

$$-(x-1)^2 \leq (x-1)^2 \sin \frac{1}{x-1} \leq (x-1)^2.$$

(i)

By Squeeze Theorem,

$$\lim_{x \rightarrow 1} (x-1)^2 \sin \frac{1}{x-1} = 0 \neq f(1).$$

Therefore, f is not continuous at $x = 1$. $f'(1)$ does not exist.

Final answer: For $x \neq 1$,

$$\begin{aligned} f'(x) &= \frac{d}{dx} \left((x-1)^2 \sin \frac{1}{x-1} \right) \\ &= 2(x-1) \sin \frac{1}{x-1} + (x-1)^2 \cdot \cos \frac{1}{x-1} \cdot \left(-\frac{1}{(x-1)^2} \right) \\ &= 2(x-1) \sin \frac{1}{x-1} - \cos \frac{1}{x-1}. \end{aligned}$$

(ii)

Thus, $f'(2) = 2(2-1) \sin 1 - \cos 1 = 2 \sin 1 - \cos 1$.

Final answer: Differentiate both sides of the equation $y^2 - 2x = 1 - 2y$ with respect to x ,

$$2y \frac{dy}{dx} - 2 = -2 \frac{dy}{dx}.$$

(iii)

Thus,

$$\frac{dy}{dx} = \frac{2}{2y+2} = \frac{1}{y+1}.$$

Final answer:

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{1}{y+1} \right) = -\frac{1}{(y+1)^2} \cdot \frac{dy}{dx} = -\frac{1}{(y+1)^2} \cdot \frac{1}{y+1} = -\frac{1}{(y+1)^3}.$$

(iv)

Question 4**(12 points)****Solution**

Final answer: Since $\frac{dv}{dt} = a(t) = \cos t$, we have $v = \sin t + C$. Given that $v(0) = 1/2$, we have $C = 1/2$. Thus,

$$\frac{ds}{dt} = v(t) = \sin t + \frac{1}{2}.$$

(i) Therefore,

$$s(t) = -\cos t + \frac{t}{2} + D.$$

Given that $s(0) = 2$, we have $D = 3$. Thus,

$$s(t) = -\cos t + \frac{t}{2} + 3.$$

Final answer: First, determine direction of motion: solve $v(t) = \sin t + \frac{1}{2} = 0$,

$$\sin t = -\frac{1}{2} \implies t = \frac{7\pi}{6}, \frac{11\pi}{6} \text{ in } [0, 2\pi].$$

Thus, $v(t) < 0$ on $(\frac{7\pi}{6}, \frac{11\pi}{6})$ and $v(t) > 0$ elsewhere on $[0, 2\pi]$. Therefore, the total distance travelled is

$$\begin{aligned} & |s(\frac{7\pi}{6}) - s(0)| + |s(\frac{11\pi}{6}) - s(\frac{7\pi}{6})| + |s(2\pi) - s(\frac{11\pi}{6})| \\ (ii) \quad &= \left(s(\frac{7\pi}{6}) - s(0)\right) + \left(s(\frac{7\pi}{6}) - s(\frac{11\pi}{6})\right) + \left(s(2\pi) - s(\frac{11\pi}{6})\right) \\ &= s(2\pi) - s(0) + 2\left(s(\frac{7\pi}{6}) - s(\frac{11\pi}{6})\right) \\ &= \pi + 2\left(-\left(-\frac{\sqrt{3}}{2}\right) + \frac{7\pi}{12} - \left(-\frac{\sqrt{3}}{2}\right) - \frac{11\pi}{12}\right) \\ &= \pi + 2\left(\sqrt{3} - \frac{4\pi}{12}\right) = \pi + 2\sqrt{3} - \frac{2\pi}{3} = \frac{\pi}{3} + 2\sqrt{3}. \end{aligned}$$

Question 5

(20 points)

Solution

Final answer:

(i)
$$a(x) = \frac{C(x)}{x} = \frac{20000}{x} + 150 + 0.5x, \quad x > 0$$

$$m(x) = \frac{dC}{dx} = 150 + x.$$

Final answer: First, find critical points of $a(x)$:

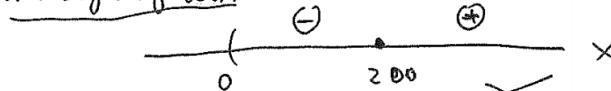
$$a'(x) = -\frac{20000}{x^2} + 0.5 = 0 \implies x^2 = 40000 \implies x = 200.$$

Second, plot critical point on x-axis and study the sign of $a'(x)$:

(ii)

x	0	200	$+\infty$
$a'(x)$	-	0	+
$a(x)$	$\searrow$	min	$\nearrow$

Or,

the sign of $a'(x)$:behavior of $a(x)$:Thus, $a(x)$ attains its minimum at $x = 200$ globally.**Final answer:** Solve $a(x) = m(x)$:

(iii)
$$\frac{20000}{x} + 150 + 0.5x = 150 + x \implies \frac{20000}{x} = 0.5x \implies x^2 = 40000 \implies x = 200.$$

Thus the intersection point is $(200, 350)$.**Final answer:** Suppose $a(x) = \frac{C(x)}{x}$ is minimized at x . Then,

(iv)
$$0 = a'(x) = \frac{d}{dx} \left(\frac{C(x)}{x} \right) = \frac{xC'(x) - C(x)}{x^2} = \frac{xm(x) - C(x)}{x^2}.$$

Thus, $a(x) = \frac{C(x)}{x} = m(x)$ at the minimizing point. QED.

Question 6**(8 points)****Solution****Final answer:**Know: $\frac{dV}{dt} = 20\pi \text{ m}^3/\text{min}$ Want: $\frac{dr}{dt}$ when $r = 1\text{m}$

$$\frac{dV}{dt} = \frac{d}{dt}(4\pi r^3/3) = 4\pi r^2 \frac{dr}{dt}.$$

$$\implies 20\pi = 4\pi(1)^2 \frac{dr}{dt} \implies \frac{dr}{dt} = 5 \text{ m/min}.$$

Question 7**(25 points)****Solution**

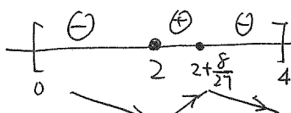
Final answer: $f'(x) = \frac{2}{3}(x-2)^{-1/3} - 1$. Set $f'(x) = 0$,

$$(i) \quad \frac{2}{3}(x-2)^{-1/3} - 1 = 0 \implies \frac{2}{3} = (x-2)^{1/3} \implies x-2 = \left(\frac{2}{3}\right)^3 = \frac{8}{27} \implies x = 2 + \frac{8}{27} = \frac{62}{27}.$$

$f'(x)$ is undefined at $x = 2$. Thus, critical points are $x = 2$ and $x = \frac{62}{27}/2 + \frac{8}{27}$.

Final answer: Plot critical points on x-axis and study the sign of $f'(x)$:

x	0	2	$\frac{62}{27}$	4	
$f'(x)$	—	0	+	0	—
$f(x)$	$\searrow$	min	$\nearrow$	max	$\searrow$

or,
 (ii) *sign of f' :*

Behavior of f :

- Decreasing on $(0, 2)$ and $(\frac{62}{27}, 4)$
- Increasing on $(2, \frac{62}{27})$

Behavior of f :

(iii) **Final answer:** By (ii), at $x = 0$ and $x = \frac{62}{27}$, f attains local maximum values.

Final answer:

$$(iv) \quad f(0) = 2^{\frac{2}{3}} + 2, \quad f\left(\frac{62}{27}\right) = \left(\frac{8}{27}\right)^{\frac{2}{3}} - \frac{8}{27} < f(0).$$

Thus the absolute maximum value of f on $[0, 4]$ is $f(0) = 2^{\frac{2}{3}} + 2$.

Final answer:

$$(v) \quad f''(x) = -\frac{2}{9}(x-2)^{-4/3} < 0, \text{ if } x \neq 2$$

Thus, f is concave down on $(0, 2)$ and $(2, 4)$.

(vi) **Final answer:** Since the concavity does not change, f has no inflection point.

Question 8**(8 points)****Solution**

Final answer: Applying Mean Value Theorem on $[0, x]$, we have

$$\sin x = \sin x - \sin 0 = \cos c(x - 0) = x \cos c, \text{ for some } c \in (0, x)$$

$$\tan x = \tan x - \tan 0 = \sec^2 d(x - 0) = x \sec^2 d, \text{ for some } d \in (0, x)$$

Since $\cos c < 1$ and $\sec^2 d > 1$ for all $c, d \in (0, x) \subset (0, \frac{\pi}{2})$, we have

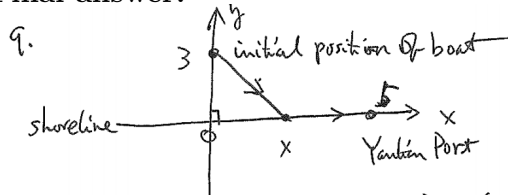
$$\sin x < x < \tan x.$$

Question 9

(8 points)

Solution

Final answer:



Let x be John's landing spot $\Rightarrow x \in [0, 5]$. Let $T(x)$ be the total time taken to reach the port if John lands at point x .

$$T(x) = \frac{\sqrt{x^2 + 9}}{3} + \frac{5 - x}{6}.$$

Want: minimize $T(x)$ on $[0, 5]$.

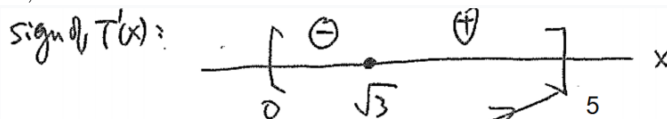
Critical point:

$$T'(x) = \frac{x}{3\sqrt{x^2 + 9}} - \frac{1}{6} = 0 \Rightarrow 2x = \sqrt{x^2 + 9} \Rightarrow 4x^2 = x^2 + 9 \Rightarrow 3x^2 = 9 \Rightarrow x = \sqrt{3}.$$

Plot critical point on x -axis and study the sign of $T'(x)$:

x	0	$\sqrt{3}$	5
$T'(x)$	-	0	+
$T(x)$		$\searrow$ min $\nearrow$	

or,



Behavior of $T(x)$:

So $T(x)$ attains its minimum at $x = \sqrt{3}$, i.e. John should land $\sqrt{3}$ km from the point on the shore to minimize his total travel time to the port.

Notes for graders: Edit the point breakdown above to match the official rubric. Add or remove solution sections as required for the actual exam.